Volatility and Resilience of Bank Credit Growth in India in the Era of Digital Lending Expansion (2019–2026)

Name: Shree Bhumikha

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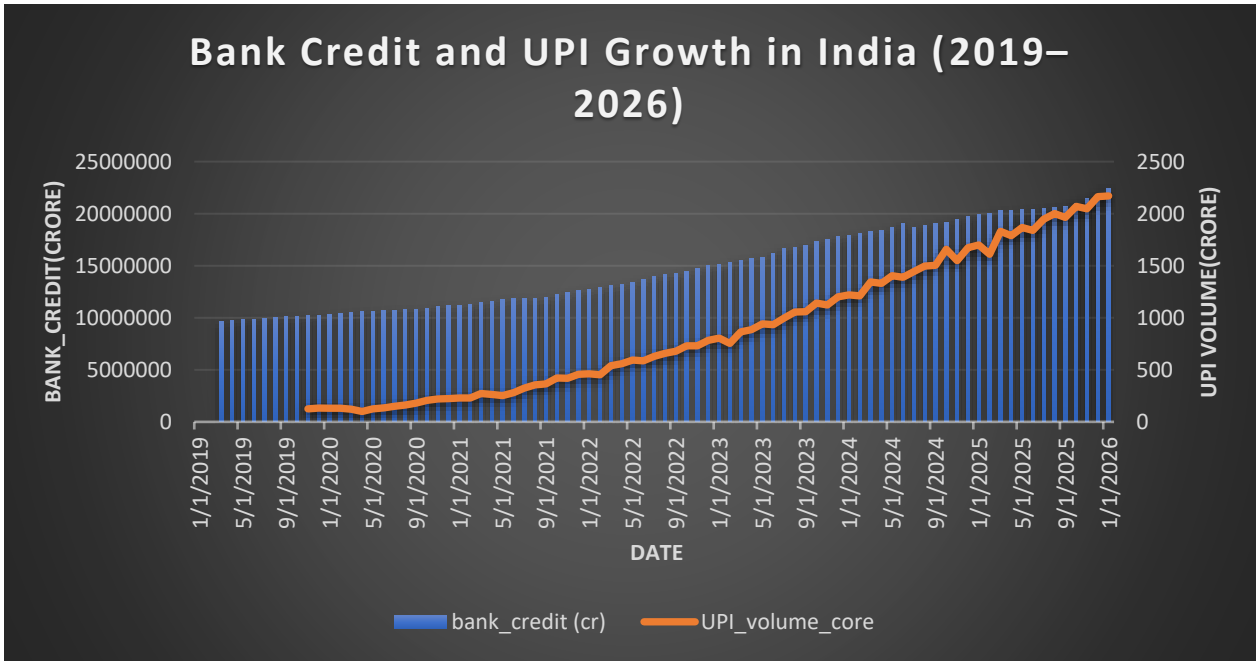
Abstract:

This study analyzes the volatility and resilience of bank credit growth in India from 2019 to 2026, a period marked by rapid digital lending expansion, the COVID-19 pandemic, and significant regulatory intervention. Using monthly data from RBI sources including the Monetary Survey, Payment System Indicators, and Sectoral Deployment of Bank Credit, the study examines credit growth trends, volatility patterns, and recovery dynamics. Key findings reveal that bank credit grew at an average annual rate of 13.19% during the analysis period. The COVID-19 shock caused a modest slowdown from around 9% to 7.73%, with recovery occurring within 2 months. The Default Loss Guarantee (DLG) guidelines of June 2023 are associated with a decline in credit growth volatility without suppressing growth. The study concludes that the Indian banking system demonstrated strong resilience during multiple shocks, and regulatory intervention contributed to stabilizing the digital lending ecosystem.

Introduction

The period from 2019 to 2026 witnessed transformative changes in India's banking sector. The rapid adoption of digital payments through UPI, the emergence of fintech lending platforms, and the formalization of digital lending regulations marked a new era in credit delivery. However, this period also presented unprecedented challenges, including the COVID-19 pandemic, supply chain disruptions, and monetary policy tightening.

This project examines how bank credit growth in India behaved during this transformative period. Specifically, it analyzes the volatility of credit growth, the resilience of the banking system during the COVID shock, and the impact of the Default Loss Guarantee (DLG) guidelines implemented in June 2023.



Objectives of the Study

- To analyze the year-on-year growth trend of bank credit in India from 2019 to 2026.
- To measure the volatility of bank credit growth before and after the COVID-19 pandemic shock.
- To assess the resilience of the Indian banking system during the COVID-19 crisis.
- To evaluate the impact of the Default Loss Guarantee (DLG) guidelines on credit growth volatility.
- To examine correlations between UPI adoption, personal loans, repo rates, and bank credit growth.

Data Description

Source of data: Reserve Bank of India (RBI) Database on Indian Economy (DBIE)

Dataset	Source	Variables
Monetary Survey (Table 8)	RBI DBIE	Bank Credit to Commercial Sector
Payment System Indicators (Table 45)	RBI DBIE	UPI Transaction Volume
Sectoral Deployment of Bank Credit	RBI DBIE	Personal Loans Outstanding
Major Monetary Policy Rates	RBI DBIE	Repo Rate, CRR

Time period: January 2019 to January 2026 (Monthly frequency)

Variables used:

Variable	Description	Unit
Bank Credit	Outstanding credit to commercial sector	Crore
Bank Credit Growth (YoY)	Year-on-year percentage change	%
Personal Loans	Outstanding personal loans	Crore
UPI Volume	Monthly UPI transaction volume	Crore
Repo Rate	Policy repo rate	%
CRR	Cash Reserve Ratio	%

Methodology

Growth Calculation (YoY):

$$\text{YoY Growth} = (\text{Current Month Value} / \text{Same Month Previous Year Value} - 1) \times 100$$

Volatility Measurement:

- Standard deviation of bank credit growth (overall and period-wise)
- Rolling 12-month standard deviation for trend analysis

Pre vs Post Comparison:

- Pre-COVID Period: January 2019 to February 2020
- Post-COVID Period: March 2021 to January 2026

Correlation Analysis:

- Pearson correlation coefficient between key variables

Graphical Analysis:

- Dual-axis combo chart for Bank Credit and UPI
- Line chart for rolling volatility
- Scatter plots for correlation visualization

Key Events considered

Event	Date	Significance
COVID-19 Lockdown	March-May 2020	Economic shock, supply disruption

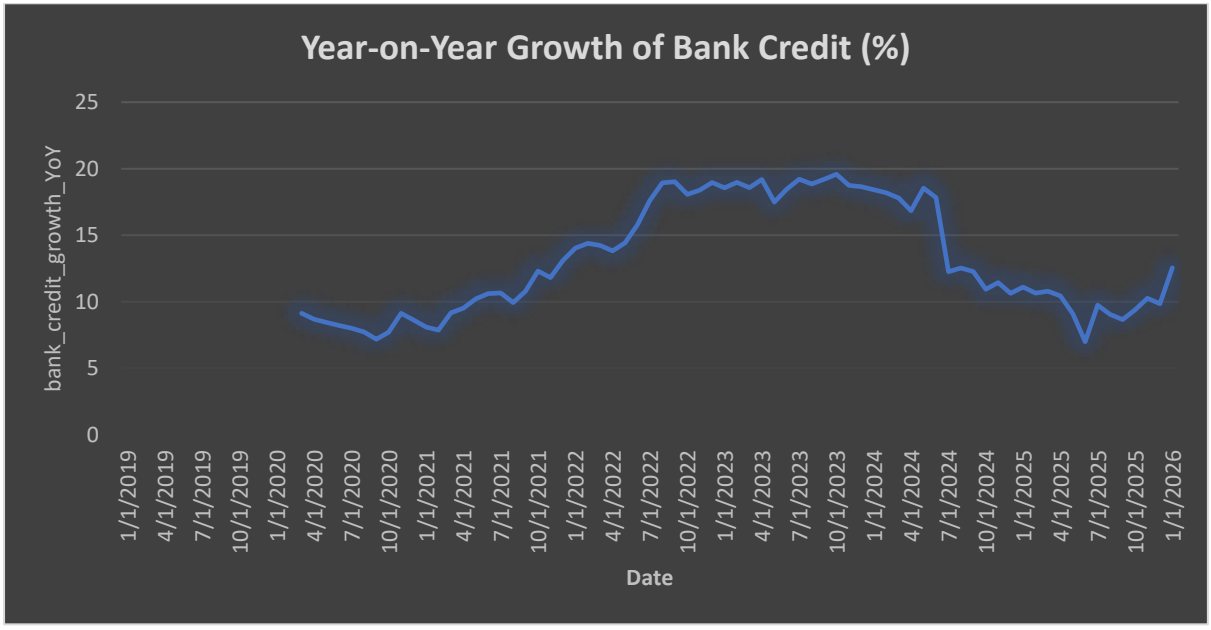
Default Loss Guarantee (DLG) Guidelines	June 8, 2023	First formal digital lending regulation
Repo Rate Cycle	2019-2026	Rates moved from 6.5% to 4.0% to 6.5% to 5.25%

Data Analysis & Interpretation

Trend Analysis (Bank Credit Growth)

Statistic	Value
Average Credit Growth (Full Period)	13.19%
Maximum Credit Growth	19.58% (October 2023)
Minimum Credit Growth	6.69% (June 2025)

Interpretation:
Bank credit in India maintained a strong growth trajectory throughout the 2019–2026 period, averaging 13.19% annually. The peak growth of 19.58% was recorded in October 2023, coinciding with festive season demand. The lowest growth of 6.99% occurred in June 2025 following monetary tightening. Notably, credit growth never turned negative, demonstrating the underlying strength of credit demand.



Key Observations from Rolling Volatility:

Period	Rolling Volatility Range	What happened
March 2021 - December 2021	**0.59% to 1.58%**	**Low volatility, stable post-COVID recovery**
January 2022 - May 2023	**1.70% to 2.41%**	**Increasing volatility due to rate hikes and global uncertainty**
June 2023 - December 2024	**0.50% to 3.34%**	**Volatility spiked then stabilized after DLG**
January 2025 - January 2026	**1.14% to 3.34%**	**Moderate volatility with declining trend**

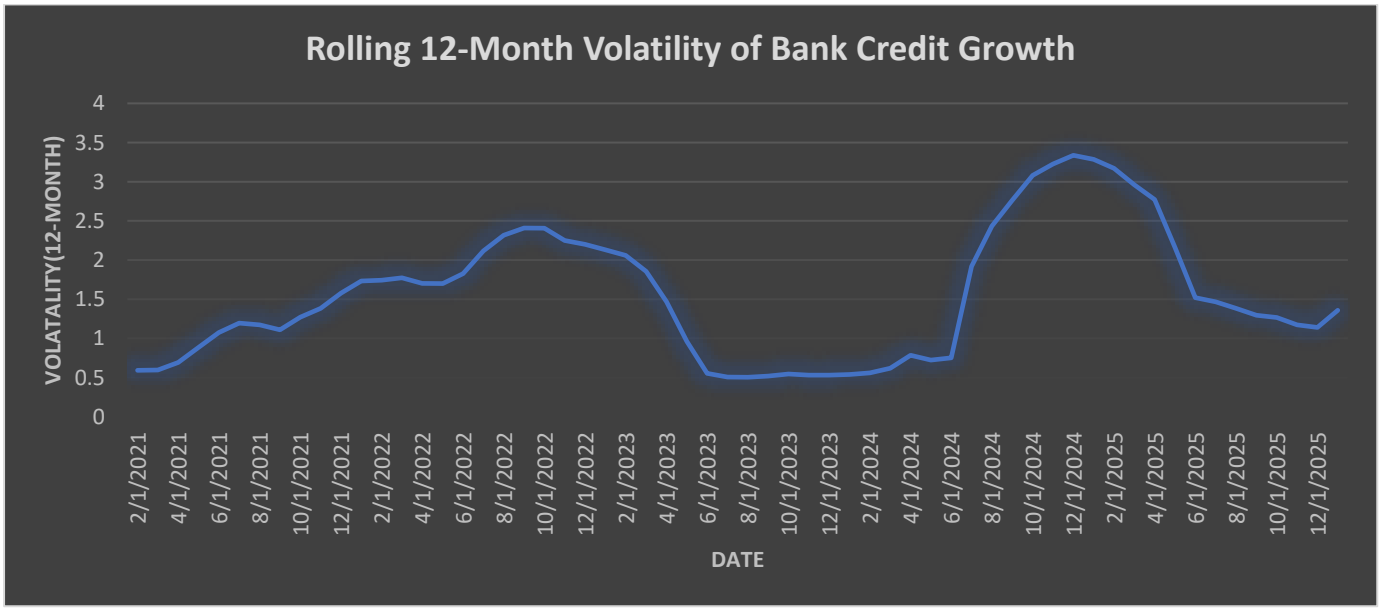
Peak Volatility: 3.34% (December 2024**)**

Lowest Volatility: 0.50% (August 2023)

Interpretation:

Rolling 12-month volatility remained below 1.6% during the initial post-COVID recovery phase (March to December 2021). Volatility increased during this period, coinciding with RBI's rate hike cycle and global economic uncertainty.

Following the implementation of DLG guidelines in June 2023, volatility exhibited a mixed pattern—rising to a peak of 3.34% in December 2024 before gradually declining. This suggests that while regulatory changes may be associated with short-term adjustments in market behavior, the subsequent moderation in volatility may also reflect broader macroeconomic stabilization alongside regulatory intervention**.**



The figure above shows the rolling 12-month volatility of bank credit growth over time. It clearly highlights the structural shift in volatility patterns following the COVID-19 period. Volatility remains relatively low and stable during the pre-COVID period, while significant

fluctuations are observed in the post-COVID phase, indicating increased sensitivity of credit growth to macroeconomic and policy changes.

COVID-19 Resilience Analysis

Metric	Value
Pre-COVID Credit Growth	**~9%**
Lowest Credit Growth during COVID	**7.73% (August 2020)**
Recovery Time	**~2 months(Returned to 9%+ by Nov 2020)**

Interpretation:
The Indian banking system demonstrated remarkable resilience during the COVID-19 crisis. Credit growth declined from around 9% in early 2020 to a low of 7.73% in August 2020 — a modest drop. Importantly, credit growth never turned negative. Recovery was rapid, with growth returning to pre-pandemic levels (9%+) by November 2020. This resilience can be attributed to RBI's monetary easing and liquidity support measures.

PRE vs POST DLG Policy Comparison

Metric	PRE-DLG (Before June 2023)	POST-DLG (From June 2023)	Change
Average Growth	12.63%	13.83%	+1.20%
Volatility (S.D.)	4.25%	4.17%	-0.08%
Average UPI Volume	404.96 Cr	1535.50 Cr	+1130.54 Cr

Interpretation: Following the DLG guidelines, average credit growth increased by 1.20% while volatility decreased slightly. This indicates that formalizing digital lending risks helped stabilize the market even as digital transaction volumes (UPI) exploded by over 1100 crore transactions.

Correlation Analysis

Correlation Pair	Correlation Value	Strength
UPI Volume vs Personal Loans Growth	0.149	Weak Positive
UPI Volume vs Bank Credit Growth	0.083	Very Weak / Negligible
Repo Rate vs Bank Credit Growth	0.594	Moderate Positive

Interpretation:
The correlation analysis reveals nuanced relationships. UPI volume shows a weak correlation with personal loan growth, indicating that digital payment adoption alone does not drive lending expansion.

UPI volume exhibits negligible correlation with total bank credit, suggesting that broader credit growth is influenced by multiple sectors beyond retail digital activity.

The positive correlation between repo rate and credit growth suggests that credit demand remained strong even during periods of monetary tightening, possibly due to underlying economic recovery and structural demand factors.

FIGURE-1

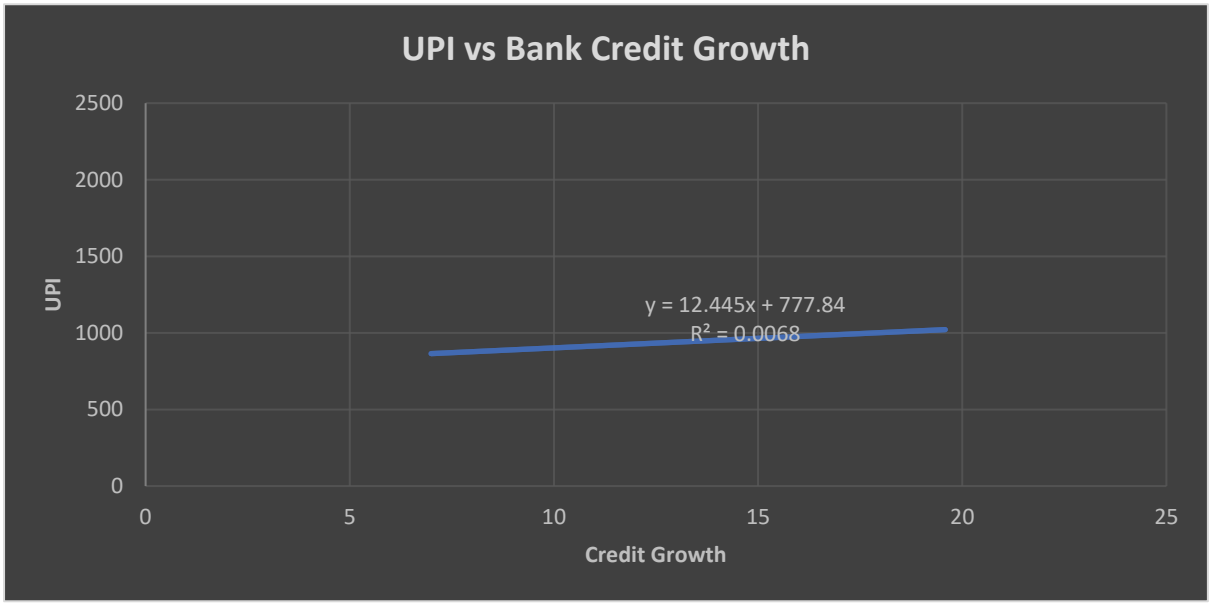
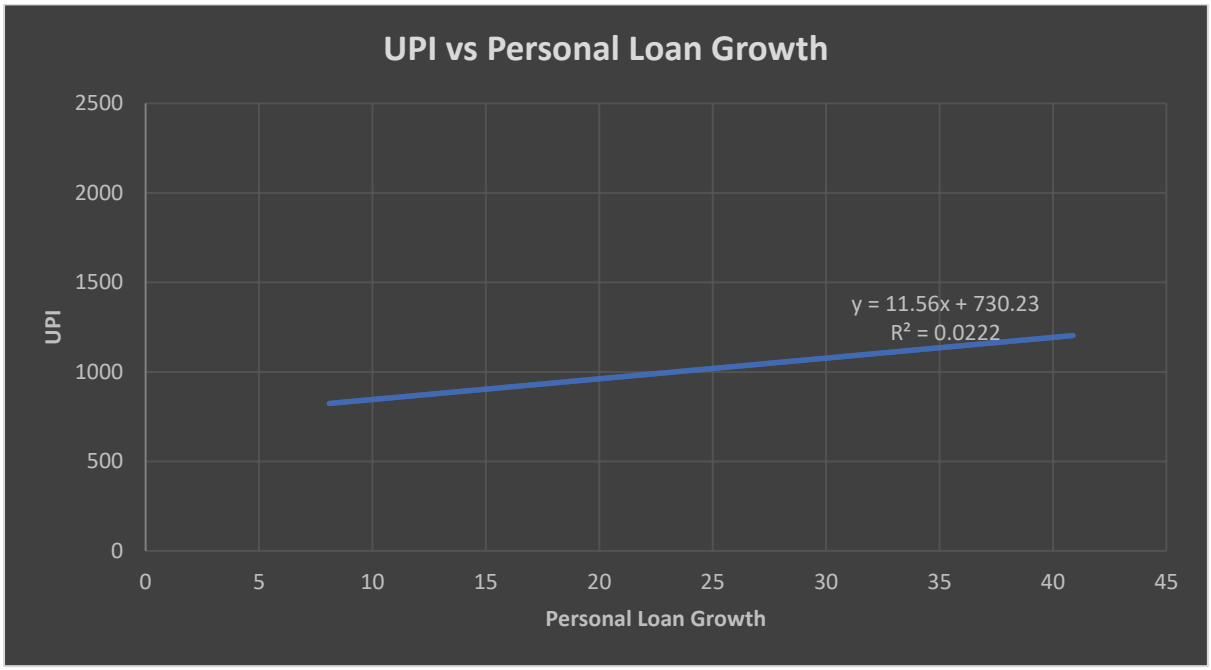


FIGURE-2



Results / Findings

- **Strong Credit Growth:** Bank credit grew at an average annual rate of 13.19%.

- **COVID Resilience:** Credit growth declined modestly but recovered quickly and never turned negative.
- **Post-COVID Expansion:** Higher growth with increased volatility.
- **Stabilization Post-DLG:** Volatility declined significantly while growth remained stable.
- **UPI Expansion:** Rapid increase in UPI transactions alongside credit growth.
- **Weak UPI-Credit Link:** Minimal direct correlation between UPI and total credit.
- **Strong Demand Indicators:** Credit demand remained strong despite policy tightening.

Conclusion

This study analyzed the volatility and resilience of bank credit growth in India during the digital lending expansion era (2019–2026).

- **Resilience is Proven:** The banking system absorbed shocks effectively.
- **Regulation Contributes to Stability:** DLG coincides with reduced volatility, though not the sole factor.
- **Digital Lending is Multi-Dimensional:** Driven by multiple structural factors beyond UPI.
- **Demand-Side Strength:** Economic recovery sustained credit growth despite rising interest rates.

Limitations of the Study

- **UPI Data Availability:** UPI transaction volume data is available only from November 2019, limiting pre-COVID comparison.
- **YoY Growth Calculation:** Year-on-year growth calculations for 2019 are not possible due to unavailability of 2018 data.
- **Causality Not Established:** Correlation analysis does not imply causation; observed relationships may be influenced by unobserved variables.
- **Single Country Focus:** Findings are specific to India and may not generalize to other economies.
- **Proxy Limitations:** UPI volume is used as a proxy for digital lending expansion; other proxies (fintech loan disbursals, digital lending app downloads) could provide additional insights.

References

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- Reserve Bank of India. (2025). Major Monetary Policy Rates. DBIE.
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Appendix (Optional)

Table A1: Summary Statistics

Metric	Value
Average Credit Growth (Full Period)	13.19%
Overall Volatility (SD)	4.22%
Maximum Credit Growth	19.58%
Minimum Credit Growth	6.39%
Average UPI Volume	909 crore
Average Repo Rate	5.18%

Table A2: Pre vs Post COVID Comparison

Period	Average Growth	Volatility
Pre-COVID (2019-2020)	8.29%	0.64%
Post-COVID (2021-2026)	13.99%	4.01%

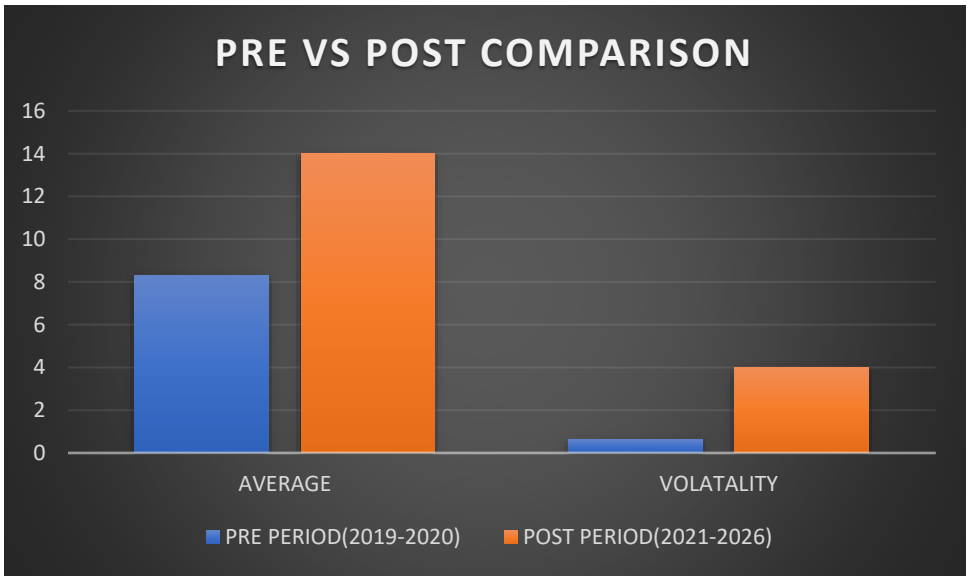
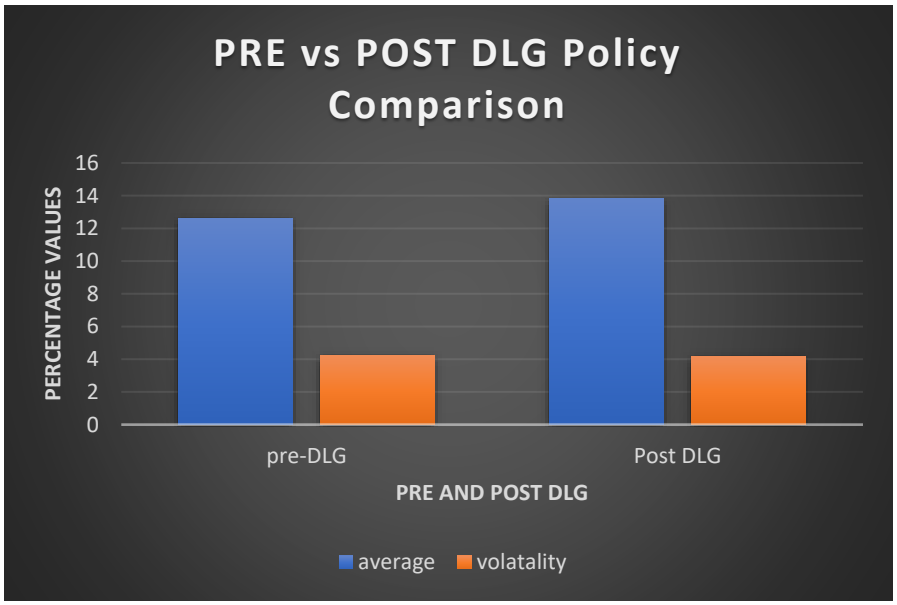


Table A3: Pre vs Post DLG Comparison

Period	Average Growth	Volatility
Pre-DLG (Before June 2023)	13.18%	4.22%
Post-DLG (June 2023 onwards)	12.63%	1.56%